

Foundations of Computing Theory—Homework 8

Due: Wednesday, February 13, 2013, 2:00pm

- You only have to explain your answer if this is stated in the question.
 - Before you start on the homework, please read the rules on collaboration and submission in the syllabus.
 - Keep in mind that, as stated in the syllabus, I will stop answering homework questions at 5pm on Tuesday.
1. Fill out the table from question 1 from the sample final questions.
 2. Draw a Venn diagram that shows the relationships between the following classes of languages:
 - (a) P.
 - (b) the CFLs.
 - (c) the decidable (recursive) languages.
 - (d) the languages whose complement is Turing-recognizable.
 - (e) the languages whose complement is decidable.
 - (f) the regular languages.
 - (g) the Turing-recognizable (recursively enumerable) languages.
 3. An unrestricted grammar is like a CFG, except that you can now have any string in $(V \cup \Sigma)^* V (V \cup \Sigma)^*$ on the left-hand side of a rule. Surprisingly, unrestricted grammars generate exactly the recursively enumerable languages. We will not show this, but we will look at an example. The following grammar generates the language $\{a^{2^i} \mid i > 0\}$.
 - $S \rightarrow ABaC$
 - $Ba \rightarrow aaB$
 - $BC \rightarrow DC$
 - $BC \rightarrow E$
 - $aD \rightarrow Da$
 - $AD \rightarrow AB$
 - $aE \rightarrow Ea$
 - $AE \rightarrow \epsilon$

Give a derivation of $aaaaaaaa$. It is ok to combine very similar steps and write $\Rightarrow^k$ when you are taking k very similar steps.

4. Consider the following problems:

$$\begin{aligned} ALL_{CFG} &= \{\langle G \rangle \mid G \text{ is a CFG and } L(G) = \Sigma^*\}. \\ \subseteq_{CFG,DFA} &= \{\langle G, A \rangle \mid G \text{ is a CFG, } A \text{ is a DFA, and } L(G) \subseteq L(A)\}. \\ \subseteq_{DFA,CFG} &= \{\langle A, G \rangle \mid A \text{ is a DFA, } G \text{ is a CFG, and } L(A) \subseteq L(G)\}. \end{aligned}$$

ALL_{CFG} is undecidable (you don't have to show this).

- (a) Show that $\subseteq_{CFG,DFA}$ is decidable. Use Sipser notation.
- (b) Use the undecidability of ALL_{CFG} to show that $\subseteq_{DFA,CFG}$ is also undecidable. Use Sipser notation.